

# **Cricket Player Performance Tracking Application : CRICKSTATS**

## **User Story:**

As a Salesforce Admin, I want to create a comprehensive player performance tracking application within Salesforce to capture and analyze detailed match-by-match information of cricket players. This application will include custom objects such as Player and Match Performance with carefully defined fields and relationships to ensure data integrity and accessibility. By leveraging master-detail and lookup relationships, the application will enable robust tracking of individual and team performances, facilitate insightful reporting and analytics, and support the overall management of player statistics, thereby enhancing strategic decision-making and player development initiatives.

## **1. INTRODUCTION**

### **1.1 Project Overview :**

CRICKSTATS is a Salesforce-based application designed to track, manage, and analyze cricket players' performance on a match-by-match basis. The system uses custom objects, relationships, dashboards, and automation tools to provide structured performance insights.

It enables coaches, analysts, and administrators to record player statistics, generate reports, and support strategic decision-making.

### **1.2 Purpose :**

The main purpose of this project is:

To digitize player performance records

To improve accuracy and accessibility of cricket statistics

To provide real-time analytics

To support player development and team planning

To automate data management using Salesforce

## 2. IDEATION PHASE

### 2.1 Problem Statement (Customer Problem Statement Template)

| <b>Problem Statement (PS)</b> | <b>I am (Customer)</b> | <b>I'm trying to</b>                       | <b>But</b>                                       | <b>Because</b>                                | <b>Which makes me feel</b>                 |
|-------------------------------|------------------------|--------------------------------------------|--------------------------------------------------|-----------------------------------------------|--------------------------------------------|
| PS-1                          | A cricket team coach   | Track player performance across matches    | I cannot get accurate statistics easily          | Data is stored in different files and formats | Frustrated and unsure about team selection |
| PS-2                          | A team manager         | Maintain complete player records           | It takes too much time to update and verify data | Most records are maintained manually          | Stressed and inefficient                   |
| PS-3                          | A sports analyst       | Analyze player trends and generate reports | I face difficulty in comparing performance s     | There is no centralized analytics system      | Confused and dissatisfied                  |

## Problem Statements

|                                                                                                                                      |                                                                                                                                      |                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
|                                                     |                                                    |                                     |
| <b>A cricket team coach</b>                                                                                                          | <b>A team manager</b>                                                                                                                | <b>A sports analyst</b>                                                                                                |
| <b>PS-1</b>                                                                                                                          | <b>PS-2</b>                                                                                                                          | <b>PS-3</b>                                                                                                            |
|  <b>A cricket team coach</b>                        |  <b>A team manager</b>                              |  <b>A sports analyst</b>            |
| <b>I am:</b> Track player performance across matches                                                                                 | <b>I am:</b> Maintain complete player records                                                                                        | <b>I am:</b> Analyze player trends and generate reports                                                                |
| <b>But:</b> I cannot get accurate statistics easily                                                                                  | <b>But:</b> It takes too much time to update and verify data                                                                         | <b>But:</b> I face difficulty in comparing performances                                                                |
| <b>Because:</b> Data is stored in different files and formats                                                                        | <b>Because:</b> Most records are maintained manually                                                                                 | <b>Because:</b> There is no centralized analytics system                                                               |
|  <b>Frustrated and unsure about team selection</b> |  <b>Frustrated and unsure about team selection</b> |  <b>Confused and dissatisfied.</b> |

## 2.2 Empathy Map Canvas

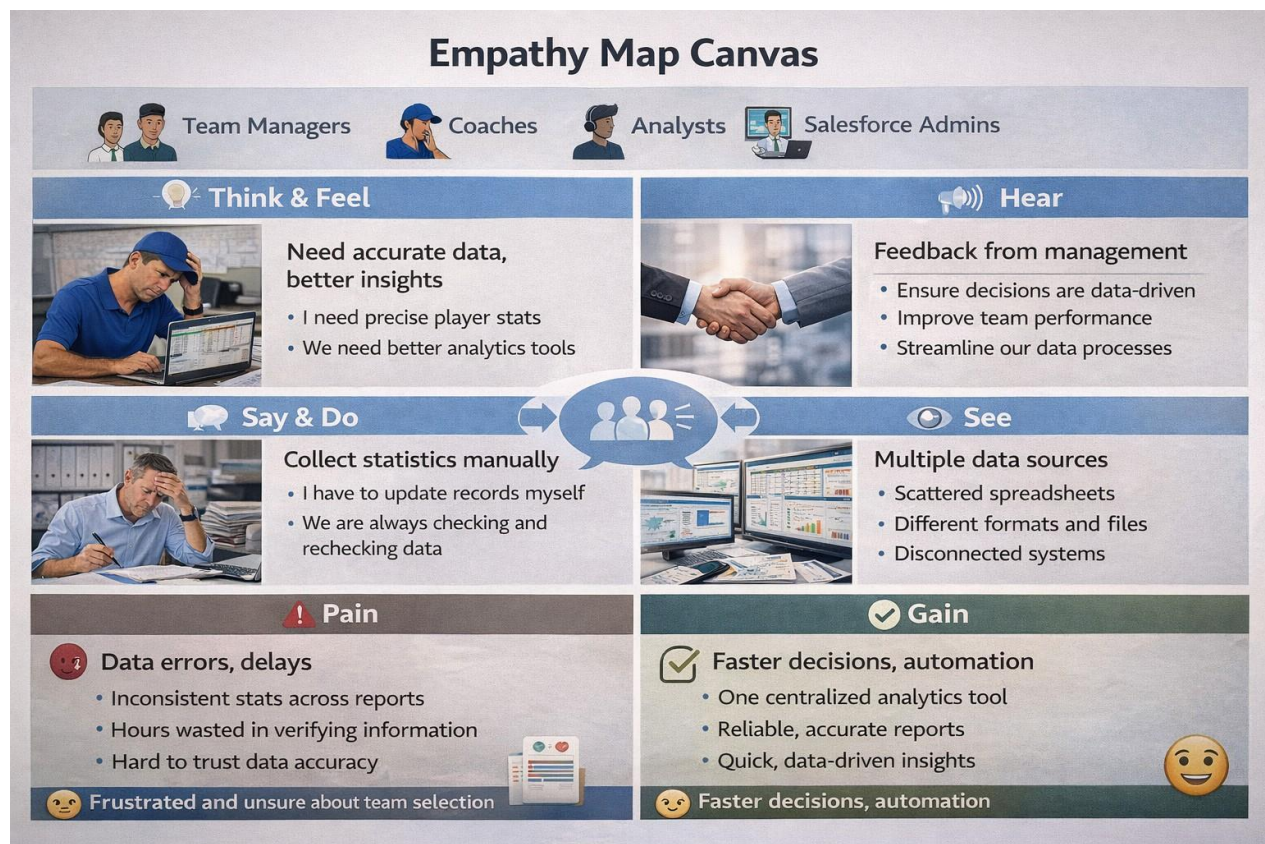
### Target Users

- Team Managers
- Coaches
- Analysts
- Salesforce Admins

### Empathy Analysis

- | Aspect                   | Description                         |
|--------------------------|-------------------------------------|
| <b>Think &amp; Feel:</b> | Need accurate data, better insights |
| <b>Hear:</b>             | Feedback from management            |
| <b>See:</b>              | Multiple data sources               |
| <b>Say &amp; Do:</b>     | Collect statistics manually         |

- **Pain:** Data errors, delays
- **Gain:** Faster decisions, automation

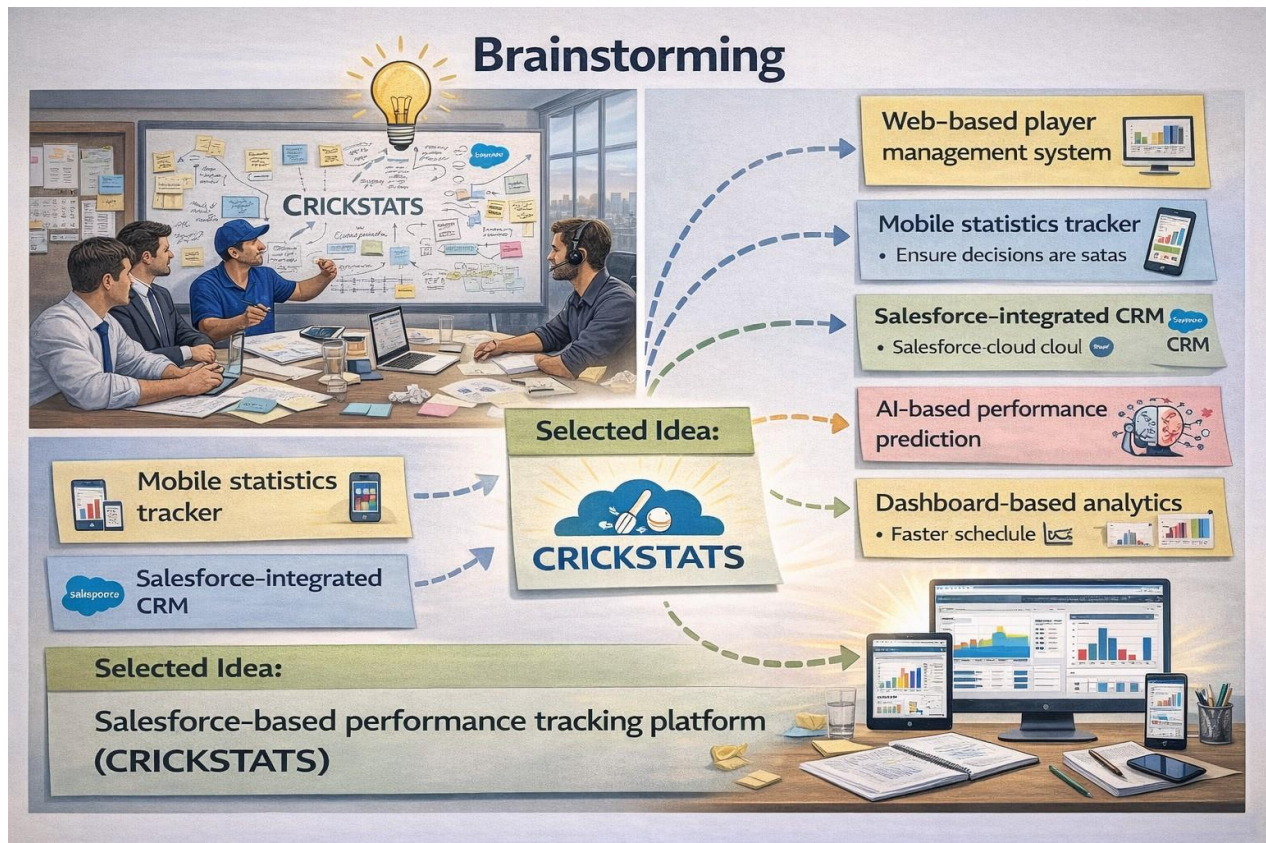


## 2.3 Brainstorming

### Brainstormed Ideas

- Web-based player management system
- Mobile statistics tracker
- Salesforce-integrated CRM
- AI-based performance prediction
- Dashboard-based analytics

**Selected Idea:** Salesforce-based performance tracking platform (CRICKSTATS)



### 3. REQUIREMENT ANALYSIS PHASE

#### 3.1 Customer Journey Map

Journey Stages:

- Login
- Enter Player Data
- Record Match Performance
- Generate Reports
- Analyze Results
- Decision Making



## 3.2 Solution Requirements

### Functional Requirements

| FR No | Requirement             |
|-------|-------------------------|
| FR-1  | Player Registration     |
| FR-2  | Match Performance Entry |
| FR-3  | Dashboard Generation    |
| FR-4  | Report Creation         |
| FR-5  | Flow Automation         |

## **Non-Functional Requirements (NFRs) – CRICKSTATS**

### **NFR-1: Usability**

The system should be easy to understand and use by coaches, managers, analysts, and admins without requiring technical training.

#### **Requirements**

- Simple and intuitive user interface
- Clear navigation menus
- Easy-to-read dashboards
- Minimal learning curve

#### **Examples:**

- A new coach should be able to:
- Log in
- Enter match data
- View player reports
- within a few minutes without special training.

Example: The Salesforce Lightning interface provides drag-and-drop dashboards and user-friendly forms.

### **NFR-2: Security**

The system must protect sensitive player data from unauthorized access.

#### **Requirements**

- Role-based access control
- Secure authentication
- Encrypted data storage
- Regular backups

.

**Examples:**

- Coaches can view player statistics
- Managers can edit records
- Analysts can generate reports
- Only admins can modify system settings

Example: Salesforce Profiles and Permission Sets restrict access to confidential information.

**NFR-3: Reliability**

The system should function correctly without failure and store data accurately.

**Requirements**

- No data loss
- Automatic backups
- Error handling
- System stability

**Example:**

If a manager enters match statistics and loses internet connection, the data should be saved automatically when reconnected.

Salesforce cloud backup ensures recovery in case of system failure.

**NFR-4: Performance**

The system should respond quickly and handle large volumes of data efficiently.

**Requirements**

- Fast page loading
- Quick report generation
- Smooth dashboard rendering
- Minimal response time

**Example:**

When an analyst generates a performance report for 100+ players, the report should load within 2–3 seconds.

Optimized Salesforce queries and caching improve response speed.

**NFR-5: Availability**

The system should be accessible whenever users need it.

**Requirements**

- 24/7 system availability
- Minimal downtime
- Cloud-based access
- Disaster recovery

**Example:**

Coaches should be able to access CRICKSTATS even during tournaments at night or on weekends.

Salesforce provides high uptime with cloud hosting.

**NFR-6: Scalability**

The system should support increasing numbers of users, players, and records.

**Requirements**

- Handle growing datasets
- Support more teams and leagues
- Expand storage capacity
- Maintain performance under load

**Example:**

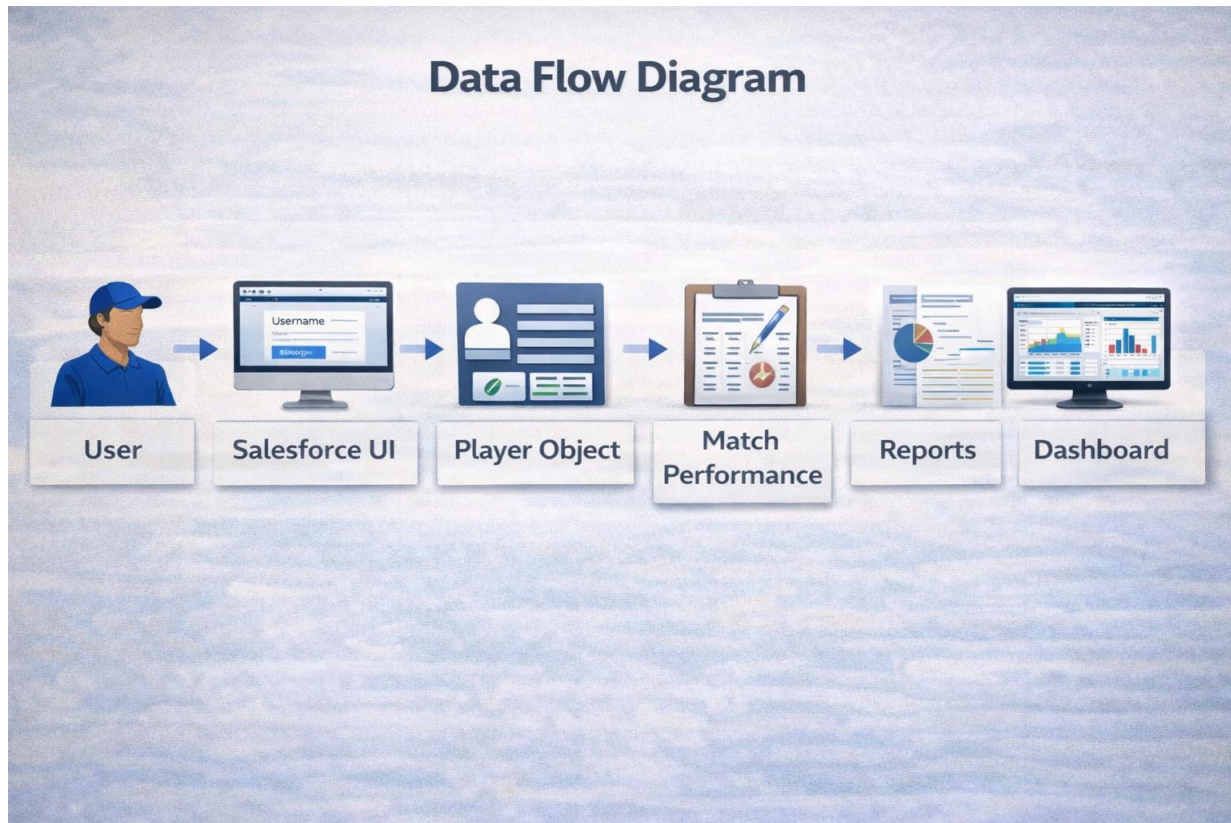
If CRICKSTATS is extended from one team to 50 teams, the system should still perform smoothly.

Salesforce cloud infrastructure automatically scales storage and processing.

### 3.3 Data Flow Diagram

#### Data Flow

User → Salesforce UI → Player Object → Match Performance → Reports → Dashboard



### 3.4 Technology Stack

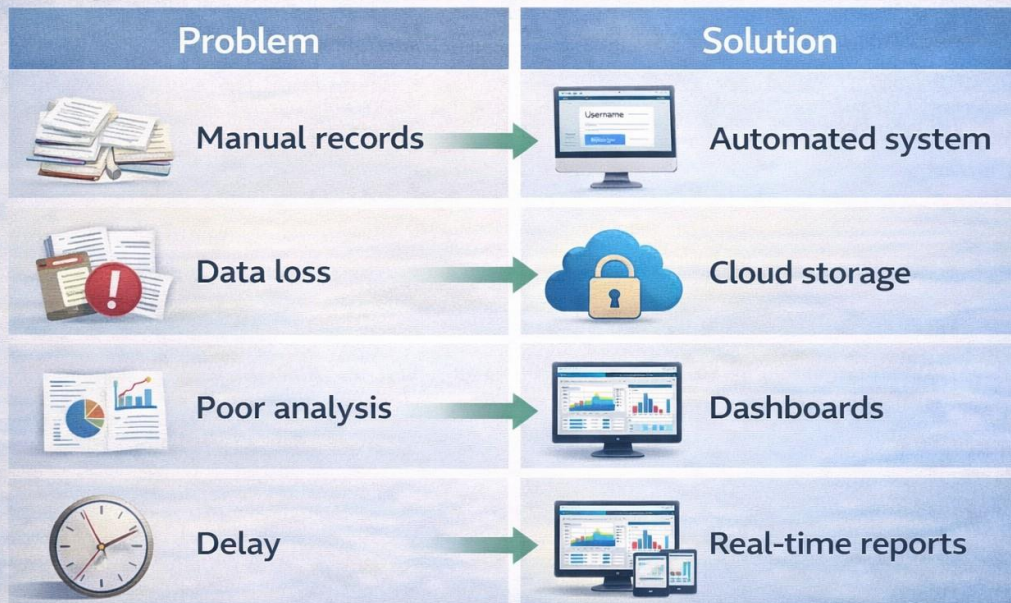
| Layer      | Technology           |
|------------|----------------------|
| Frontend   | Salesforce Lightning |
| Analytics  | Reports & Dashboards |
| Automation | Salesforce Flows     |
| Backend    | Salesforce Objects   |
| Database   | Salesforce Cloud     |

## 4. PROJECT DESIGN PHASE

### 4.1 Problem–Solution Fit

The CRICKSTATS system effectively addresses the key challenges faced in traditional cricket performance management. Earlier, most player and match data were maintained using manual records, which were time-consuming and prone to errors. This issue has been resolved by implementing an automated system through Salesforce, which ensures accurate and efficient data entry and management. The problem of data loss, caused by improper storage and lack of backups, is overcome by using secure cloud storage that safely stores all information and allows easy retrieval. Poor analysis of player performance, due to scattered and unorganized data, is addressed through interactive dashboards that present clear visual insights and reports. Additionally, delays in accessing performance information and generating reports are eliminated by providing real-time reporting features, enabling users to make faster and more informed decisions. Thus, CRICKSTATS successfully bridges the gap between existing problems and modern technological solutions.

## Problem-Solution Fit



## 4.2 Proposed Solution

### Description

**CRICKSTATS provides:**

- Custom objects (Player, Match Performance)
- Master-detail relationships
- Roll-up summaries
- Lightning App
- Automated flows
- Interactive dashboards

### Key Features

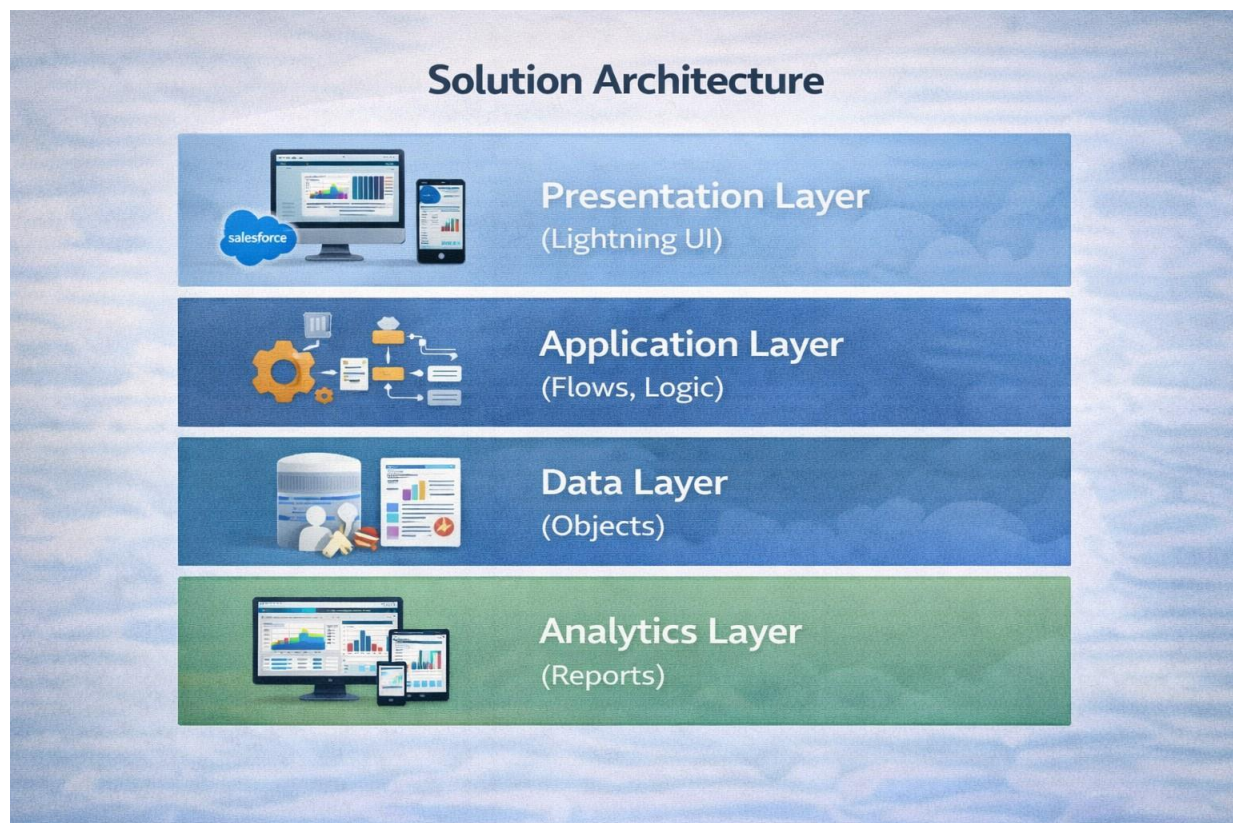
- Centralized database
- Live statistics
- Secure access

- Scalable design
- Easy reporting

## 4.3 Solution Architecture

### Architecture Layers

- Presentation Layer (Lightning UI)
- Application Layer (Flows, Logic)
- Data Layer (Objects)
- Analytics Layer (Reports)



## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

- Agile-based execution
- Object creation → automation → reporting
- Sprint-wise development

#### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                              | Story Points | Priority | Team Members |
|----------|-------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------|----------|--------------|
| Sprint-1 | Developer Setup               | USN-1             | As an admin, I want to create a Salesforce developer account so that I can configure and deploy the CRICKSTATS system.         | 1            | High     | Member1      |
| Sprint-1 | Data Modeling                 | USN-2             | As an admin, I want to create Player and Match Performance objects so that player and match data can be stored systematically. | 3            | High     | Member2      |
| Sprint-2 | Navigation                    | USN-3             | As a user, I want separate tabs for each object so that I can easily navigate and manage records within the application.       | 3            | High     | Member3      |
| Sprint-2 | UI Design                     | USN-4             | As a coach, I want a dedicated Lightning App so that all performance tracking features are accessible from one dashboard.      | 3            | High     | Member4      |
| Sprint-2 | Data Structure                | USN-5             | As an admin, I want to create player and match fields so that complete performance details can be captured accurately.         | 5            | Medium   | Member1      |
| Sprint-2 | Data Structure                | USN-6             | As a user, I want customized page layouts so that only relevant fields are displayed during data entry.                        | 5            | High     | Member2      |
| Sprint-2 | Automation                    | USN-7             | As a system, I want to validate player and match inputs so that incorrect data entries are prevented.                          | 3            | High     | Member4      |
| Sprint-3 | Automation                    | USN-8             | As a system, I want automated flows to update player statistics so that performance data remains accurate.                     | 5            | High     | Member1      |

### Project Tracker, Velocity & Burndown Chart: (4 Marks)

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint-1 | 20                 | 6 Days   | 24 Oct 2022       | 29 Oct 2022               | 20                                              | 29 Oct 2022                  |
| Sprint-2 | 20                 | 6 Days   | 31 Oct 2022       | 05 Nov 2022               |                                                 |                              |
| Sprint-3 | 20                 | 6 Days   | 07 Nov 2022       | 12 Nov 2022               |                                                 |                              |
| Sprint-4 | 20                 | 6 Days   | 14 Nov 2022       | 19 Nov 2022               |                                                 |                              |
|          |                    |          |                   |                           |                                                 |                              |
|          |                    |          |                   |                           |                                                 |                              |
|          |                    |          |                   |                           |                                                 |                              |
|          |                    |          |                   |                           |                                                 |                              |

### 6. Project Development Phase:

#### Project Flow

Milestone 1 : Creation of developer account

Milestone 2 : Object Creation

Milestone 3 : Tabs

Milestone 4 : The Lightning App

Milestone 5 : Fields & Relationships

Milestone 6 : Roll-up Summary

Milestone 7 : Flows

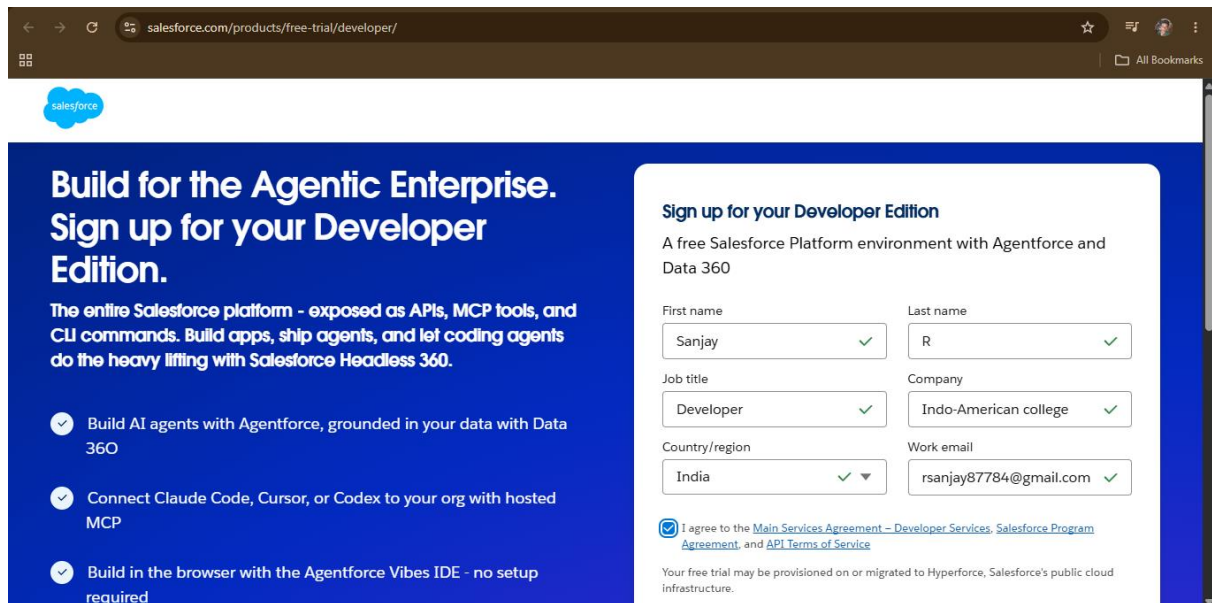
#### Implementation

#### Milestone 1 - Salesforce developer account creation :

## Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to <https://developer.salesforce.com/signup>
2. On the sign up form, enter the following details :



The screenshot shows the Salesforce Developer Edition sign-up page. On the left, a blue banner reads "Build for the Agentic Enterprise. Sign up for your Developer Edition." and lists features like building AI agents, connecting Claude Code, and using the Agentforce Vibes IDE. On the right, a white form titled "Sign up for your Developer Edition" contains the following fields: First name (Sanjay), Last name (R), Job title (Developer), Company (Indo-American college), Country/region (India), and Work email (rsanjay87784@gmail.com). Each field has a green checkmark indicating it is valid. Below the form, there is a checkbox for agreeing to the terms of service and a note about the free trial being provisioned on Hyperforce infrastructure.

- 1) First name & Last name
- 2) Email
- 3) Role : Developer
- 4) Company : College or Company Name
- 5) County : India
- 6) Postal Code : pin code
- 7) Username : should be a combination of your name and company

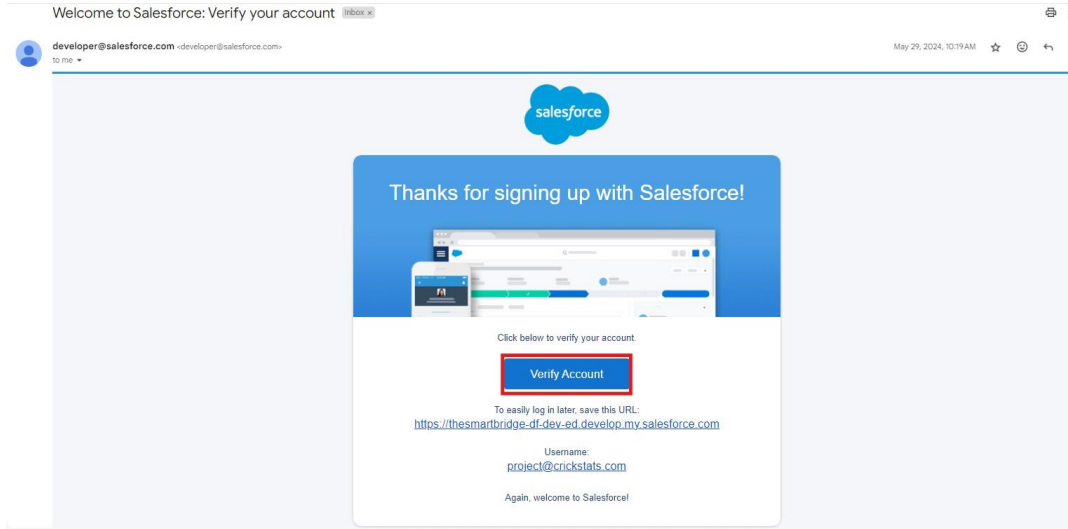
This need not be an actual email id, you can give anything in that format :

[username@organization.com](mailto:username@organization.com)

Click on sign me up after filling these.

## Activity 2: Account Activation :

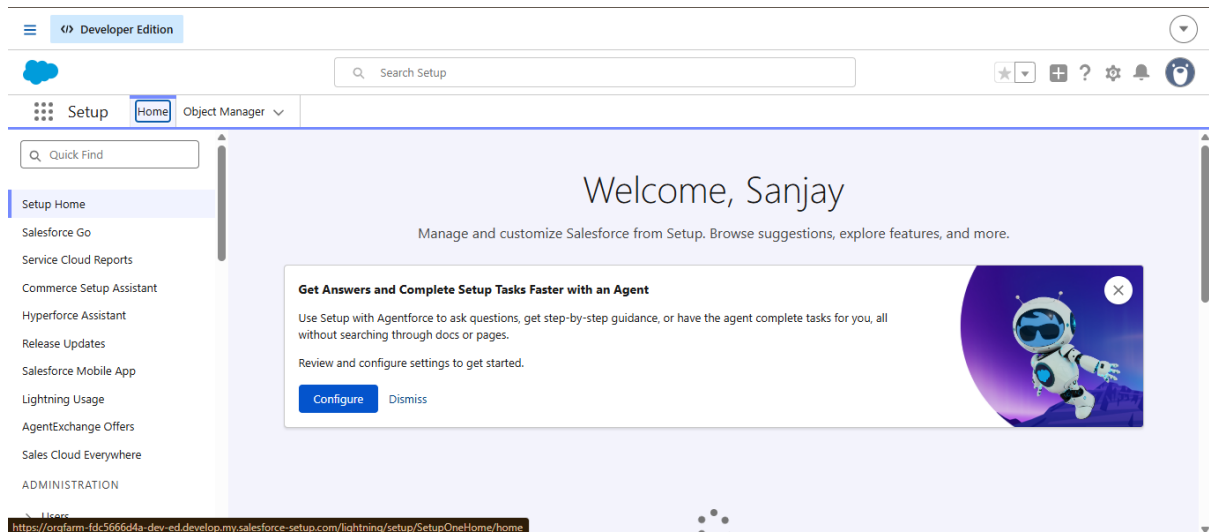
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.
2. Click on Verify Account.



3. Give a password and answer a security question and click on change password.

This is a screenshot of the "Change Your Password" page in Salesforce. The page title is "Change Your Password". Below the title, it says "Enter a new password for lead@sb.com. Make sure to include at least:". There are three green checkmarks indicating requirements: "8 characters", "1 letter", and "1 number". Below these are three input fields, each with a red border. The first is labeled "\* New Password" and has a "Good" status. The second is labeled "\* Confirm New Password" and has a "Match" status. The third is labeled "Security Question" and has a dropdown menu showing "In what city were you born?". Below the security question is an "Answer" field with the text "asdfghjkl". At the bottom of the form is a blue button labeled "Change Password".

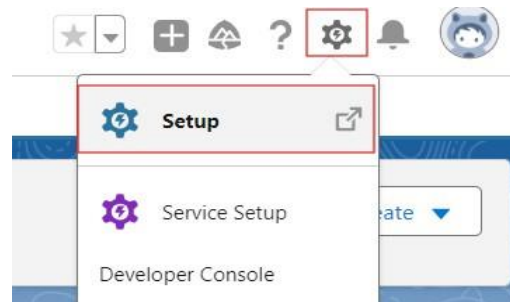
4. Then you will redirect to your salesforce setup page.



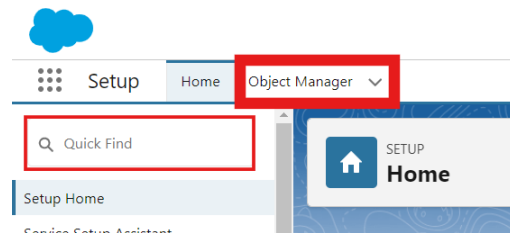
## Milestone 2 - Object Creation

### Activity 1 : Player Object

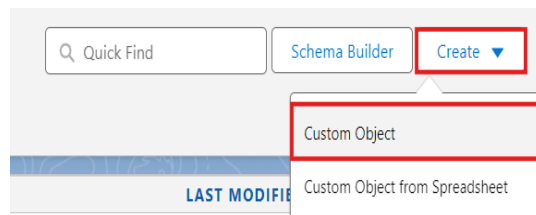
1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.



2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.



3. On the Object Manager page, click Create | Custom Object.



1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
  - **Label** : Player
  - **Plural Label** : Players
  - Enter Record Name Label and Format  
Record Name : Player Name  
Data Type : Text
  - **In Optional features** : check the boxes for Allow Reports | Allow Activities | Track Field History.
  - **Search Status** : check the box for Allow Search.
  - Save

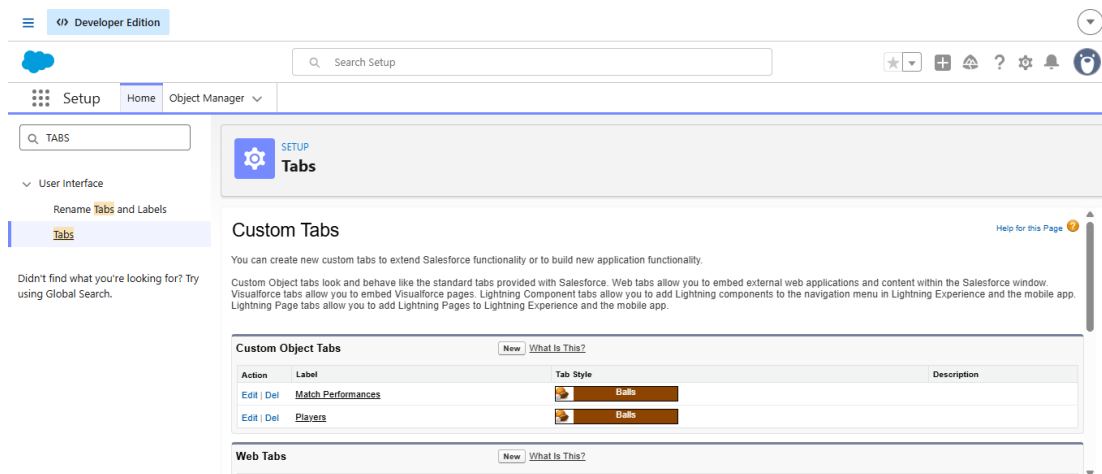
## Activity 2 : Match Performance Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
  - **Label** : Match Performance
  - **Plural Label** : Match Performances
  - Enter Record Name Label and Format  
Record Name : Customer Account Name  
Data Type : Text
  - **In Optional features** : check the boxes for Allow Reports | Allow Activities | Track Field History.
  - **Search Status** : check the box for Allow Search.
  - Save.

## Milestone 3 - Tabs

### Activity 1 : Custom tabs creation

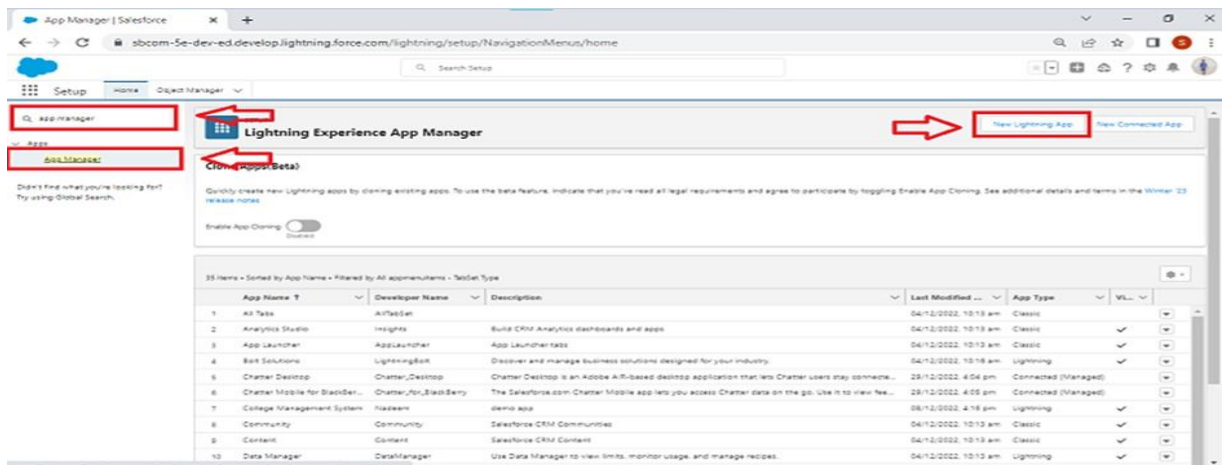
1. Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)
2. Select Object(Player) → Select the tab style → Next (Add to profiles page) keep it as default → Next (Add to Custom App) uncheck the include tab .
3. Make sure that the Append tab to users' existing personal customizations is checked.
4. Click save.
5. Repeat the same steps for Match Performances.



## Milestone 4 - The Lightning App

### Activity 1 : Create a lightning app

1. Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.



2. Fill the app name in app details and branding as follow  
 App Name : CRICKSTATS  
 Developer Name : Auto Populates  
 Add image optional (if you want to give any image you can otherwise not mandatory) - Add Primary color Hex or leave it to default.  
 Then click Next.
3. In App options  
 Navigation style : Standard navigation  
 Setup experience : setup  
 Supporters form factors : Desktop and phone  
 Then click Next
4. In Utility items  
 Utility Bar alignment : Default  
 Then click Next.
5. Navigation items  
 Select the created Custom Objects and required standard objects
  - Players
  - Match Performances
  - Reports
  - Dashboards

## Navigation Items

Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename navigation items. Navigation items are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.

The screenshot shows the 'Navigation Items' configuration interface. It consists of two main panels: 'Available Items' on the left and 'Selected Items' on the right. The 'Available Items' panel has a search bar with the text 'Dash' and a list of items below it, with 'Dashboards' highlighted by a red box. The 'Selected Items' panel has a list of items: 'Players', 'Match Performances', and 'Reports', all of which are highlighted by a red box. Between the two panels are navigation arrows: a right arrow (highlighted with a red box) and a left arrow. At the bottom of the interface are 'Cancel' and 'Save' buttons, with the 'Save' button highlighted by a red box.

## 6. To Add User Profiles : System Administrator

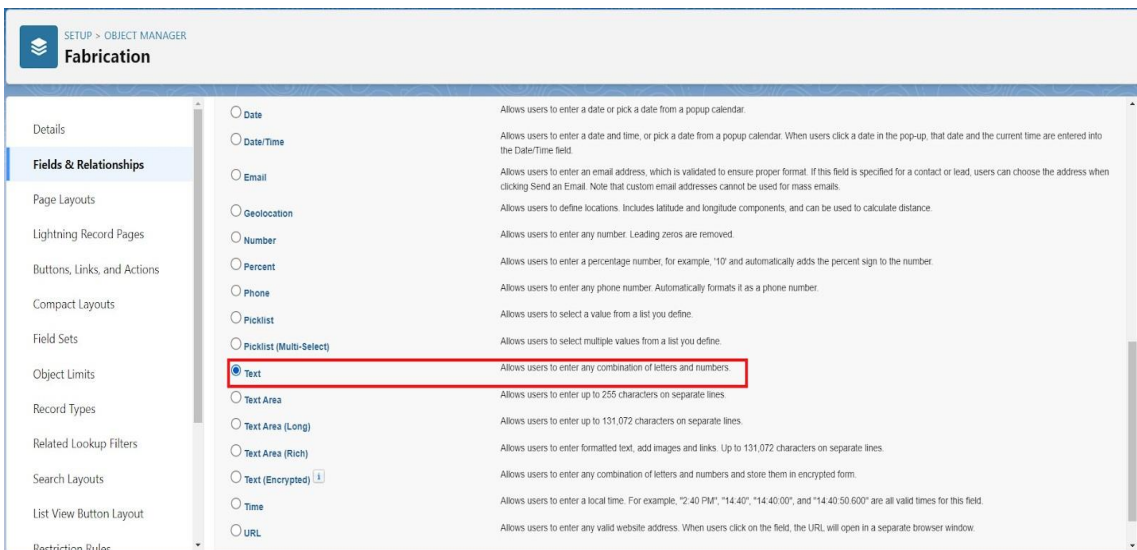
The screenshot shows the 'New Lightning App' configuration interface, specifically the 'User Profiles' section. The title 'New Lightning App' is at the top. Below it, the section is titled 'User Profiles' with the instruction 'Choose the user profiles that can access this app.' There are two panels: 'Available Profiles' on the left and 'Selected Profiles' on the right. The 'Available Profiles' panel has a search bar with the text 'System administrator' and a list of items below it, with 'System administrator' highlighted by a red box. The 'Selected Profiles' panel has a list of items: 'System Administrator', which is also highlighted by a red box. Between the two panels are navigation arrows: a right arrow (highlighted with a red box) and a left arrow. At the bottom of the interface is a progress bar with four steps, the last of which is active. To the right of the progress bar is a 'Save & Finish' button, which is highlighted by a red box and has a red arrow pointing to it.

## 7. Click Save & Finish.

## Milestone 5 : Fields & Relationships

### Activity 1 : Create Custom fields for Player object

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist.
6. Next, choose a data type as Text. Choosing a data type helps you format the field input.



7. Click on next.
8. Data Type : **Text**  
Field Label: Player Name  
Field Name : Auto Populates  
Length : 50  
Click on Next → Next → Save and new.

9. Repeat the Same steps for remaining fields

- Data Type : **Text**  
Field Label: Player Number  
Field Name : Auto Populates  
Length : 3  
Click on Next → Next → Save and new.
- Data Type : **Rich Text Area**  
Field Label : Profile Picture  
Field Name : Auto Populates  
Click on Next → Next → Save and new.
- Data Type : **Date**  
Field Label : Date of Birth  
Field Name : Auto Populates  
Click on Next → Next → Save and new.
- Data Type : **Picklist**  
Field Label : Role  
Field Name : Auto Populates  
Value : Enter value, with each value separated by a new line.  
Batsman  
Bowler  
Batsman & Wicket Keeper  
All Rounder
- Data Type : **Picklist**  
Field Label : Batting Style  
Field Name : Auto Populates  
Value : Enter value, with each value separated by a new line.  
Right- Handed  
Left - Handed  
Click on Next → Next → Save and new.
- Data Type : **Picklist**  
Field Label : Bowling Style  
Field Name : Auto Populates  
Value : Enter value, with each value separated by a new line.  
Right-Arm Fast  
Left-Arm Fast

Right-Arm Medium  
Left-Arm Medium  
Right-Arm Spin  
Left-Arm Spin

Click on Next → Next → Save and new.

- Data Type : **Picklist**

Field Label : Nationality

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line.

Australia  
Bangladesh  
England  
India  
New Zealand  
Pakistan  
South Africa  
Sri Lanka  
West Indies  
Afghanistan  
Zimbabwe  
Ireland  
Scotland  
Netherlands  
United Arab Emirates  
Nepal  
United States of America

Click on Next → Next → Save and new.

- Data Type : **Picklist**

Field Label : Team

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line

Chennai Super Kings (CSK)  
Delhi Capitals (DC)  
Kolkata Knight Riders (KKR)  
Mumbai Indians (MI)  
Punjab Kings (PBKS)  
Rajasthan Royals (RR)  
Royal Challengers Bangalore (RCB)  
Sunrisers Hyderabad (SRH)  
Gujarat Titans (GT)  
Lucknow SuperGiants (LSG)

- Data Type : **Formula**  
Field Label : AVG Score  
Field Name : Auto Populates  
Formula Return : **Number**  
Formula : IF(  

Matches\_Played\_\_c > 0,

Total\_Runs\_\_c / Matches\_Played\_\_c ,

0

)

The screenshot shows the Salesforce formula editor interface. At the top, there are two tabs: 'Simple Formula' and 'Advanced Formula'. Below the tabs, there are two buttons: 'Insert Field' and 'Insert Operator'. The main text area contains the formula: 'AVG Score (Number) = IF ( Matches\_Played\_\_c > 0, Total\_Runs\_\_c / Matches\_Played\_\_c , 0 )'. At the bottom, there is a 'Check Syntax' button and a green message: 'No syntax errors in merge fields or functions. (Compiled size: 130 characters)'.

Click on Next → Next → Save & New.

- Data Type : **Lookup Relationship**  
Related To : Match Performance  
Field Label : Match Performance  
Field Name : Auto Populates  
Click on Next → Next → Save.

## Activity 2 : Create Custom fields for Match Performance object.

1. Data Type : **Text**  
Field Label: Match Performance Name  
Field Name : Auto Populates  
Length : 80  
Click on Next → Next → Save and new.
2. Data Type : **Checkbox**  
Field Label: Player of the Match

Field Name : Auto Populates  
Click on Next → Next → Save and new.

3. Data Type : **Picklist**

Field Label: Match Result

Field Name : Auto Populates

Value : Enter value, with each value separated by a new line

WON

LOST

DRAW

Click on Next → Next → Save and new.

4. Data Type : **Number**

Field Label: Runs Scored

Field Name : Auto Populates

Click on Next → Next → Save and new.

5. Data Type : **Number**

Field Label: Balls Faced

Field Name : Auto Populates

Click on Next → Next → Save and new.

6. Data Type : **Number**

Field Label: Fours

Field Name : Auto Populates

Click on Next → Next → Save and new.

7. Data Type : **Number**

Field Label: Sixes

Field Name : Auto Populates

Click on Next → Next → Save and new.

8. Data Type : **Number**

Field Label: Overs Bowled

Field Name : Auto Populates

Click on Next → Next → Save and new.

9. Data Type : **Number**

Field Label: Wickets Taken

Field Name : Auto Populates

Click on Next → Next → Save and new.

10. Data Type : **Number**  
Field Label: Runs Conceded  
Field Name : Auto Populates  
Click on Next → Next → Save and new.
11. Data Type : **Number**  
Field Label: Maiden Overs  
Field Name : Auto Populates  
Click on Next → Next → Save and new.
12. Data Type : **Number**  
Field Label: Stumpings  
Field Name : Auto Populates  
Click on Next → Next → Save and new
13. Data Type : **Number**  
Field Label: Catches  
Field Name : Auto Populates  
Click on Next → Next → Save and new.
14. Data Type : **Master - Detail Relationship**  
Related To : Player  
Field Label : Player  
Field Name : Auto Populates  
Click on Next → Next → Save.

## **Milestone 6 : Roll-up Summary**

### **Activity 1 : Create Roll-up Summary for Player object.**

**Note** :- Roll-up Summary can only be created on parent object in Master- Detail Relationship.

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Player**

- From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
- Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist. Click Next,
- Choose a data type as **Roll-up Summary**. Click Next.

Field Label: Matches Played

Field Name : Auto Populates

Summarized Object : Match Performances

Select Roll-up Type : COUNT

Select Object to Summarize

Master Object: Player

Summarized Object: Match Performances

Select Roll-Up Type

☒ COUNT

☐ SUM

☐ MIN

☐ MAX

Field to Aggregate: --None--

Filter Criteria

☒ All records should be included in the calculation

☐ Only records meeting certain criteria should be included in the calculation

Previous Next Cancel

Filter Criteria : All records should be included in the calculation.

Click on Next → Next → Save and new.

- Data Type : Roll-up Summary
- Field Label: Number of 50's
- Field Name : Auto Populates
- Summarized Object : Match Performances
- Select Roll-up Type : COUNT
- Filter Criteria : Only records meeting certain criteria should be included in the calculation.
- Click on Next → Next → Save and new.

Select Roll-Up Type

☒ COUNT

☐ SUM

☐ MIN

☐ MAX

Field to Aggregate: --None--

Filter Criteria

☐ All records should be included in the calculation

☒ Only records meeting certain criteria should be included in the calculation

| Field       | Operator         | Value |     |
|-------------|------------------|-------|-----|
| Runs Scored | greater or equal | 50    | AND |
| Runs Scored | less than        | 100   | AND |
| --None--    | --None--         |       | AND |
| --None--    | --None--         |       | AND |
| --None--    | --None--         |       |     |

For checkbox fields, enter a value of True for checked or False for not checked. For picklist fields, enter the master picklist field value in your corporate language.

Save Cancel

- Data Type : Roll-up Summary  
Field Label: Number of 100's  
Field Name : Auto Populates  
Summarized Object : Match Performances  
Select Roll-up Type : COUNT  
Filter Criteria : Only records meeting certain criteria should be included in the calculation.  
Click on Next → Next → Save and new.

Master Object: Player  
Summarized Object: Match Performances

**Select Roll-Up Type**

☒ COUNT  
☐ SUM  
☐ MIN  
☐ MAX

Field to Aggregate: --None--

**Filter Criteria**

☐ All records should be included in the calculation  
☒ Only records meeting certain criteria should be included in the calculation

| Field       | Operator         | Value |     |
|-------------|------------------|-------|-----|
| Runs Scored | greater or equal | 100   | AND |
| Runs Scored | less than        | 200   | AND |
| --None--    | --None--         |       | AND |
| --None--    | --None--         |       | AND |
| --None--    | --None--         |       |     |

For checkbox fields, enter a value of True for checked or False for not checked. For picklist fields, enter the master picklist field value in your corporate language.

Previous Next Cancel

- Data Type : Roll-up Summary  
Field Label: Total 4's  
Field Name : Auto Populates  
Summarized Object : Match Performances  
Select Roll-up Type : SUM  
Field To Aggregate : Fours  
Filter Criteria : All records should be included in the calculation.  
Click on Next → Next → Save and new.
- Data Type : Roll-up Summary  
Field Label: Total 6's  
Field Name : Auto Populates  
Summarized Object : Match Performances  
Select Roll-up Type : SUM  
Field To Aggregate : Sixes  
Filter Criteria : All records should be included in the calculation.  
Click on Next → Next → Save and new.
- Data Type : Roll-up Summary  
Field Label: Total Runs  
Field Name : Auto Populates

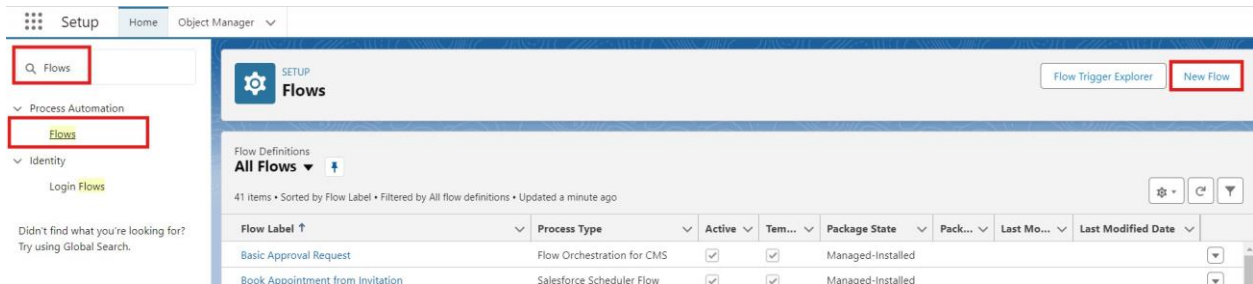
Summarized Object : Match Performances  
Select Roll-up Type : SUM  
Field To Aggregate : Runs Scored  
Filter Criteria : All records should be included in the calculation.  
Click on Next → Next → Save and new.

- Data Type : Roll-up Summary  
Field Label: Total Overs  
Field Name : Auto Populates  
Summarized Object : Match Performances  
Select Roll-up Type : SUM  
Field To Aggregate : Overs Bowled  
Filter Criteria : All records should be included in the calculation.  
Click on Next → Next → Save and new.
- Data Type : Roll-up Summary  
Field Label: Total Wickets  
Field Name : Auto Populates  
Summarized Object : Match Performances  
Select Roll-up Type : SUM  
Field To Aggregate : Wickets Taken  
Filter Criteria : All records should be included in the calculation.  
Click on Next → Next → Save and new.
- Data Type : Roll-up Summary  
Field Label: Total Stumpings  
Field Name : Auto Populates  
Summarized Object : Match Performances  
Select Roll-up Type : SUM  
Field To Aggregate : Stumpings  
Filter Criteria : All records should be included in the calculation.  
Click on Next → Next → Save and new.
- Data Type : Roll-up Summary  
Field Label: Total Catches  
Field Name : Auto Populates  
Summarized Object : Match Performances  
Select Roll-up Type : SUM  
Field To Aggregate : Catches  
Filter Criteria : All records should be included in the calculation.  
Click on Next → Next → Save and new.

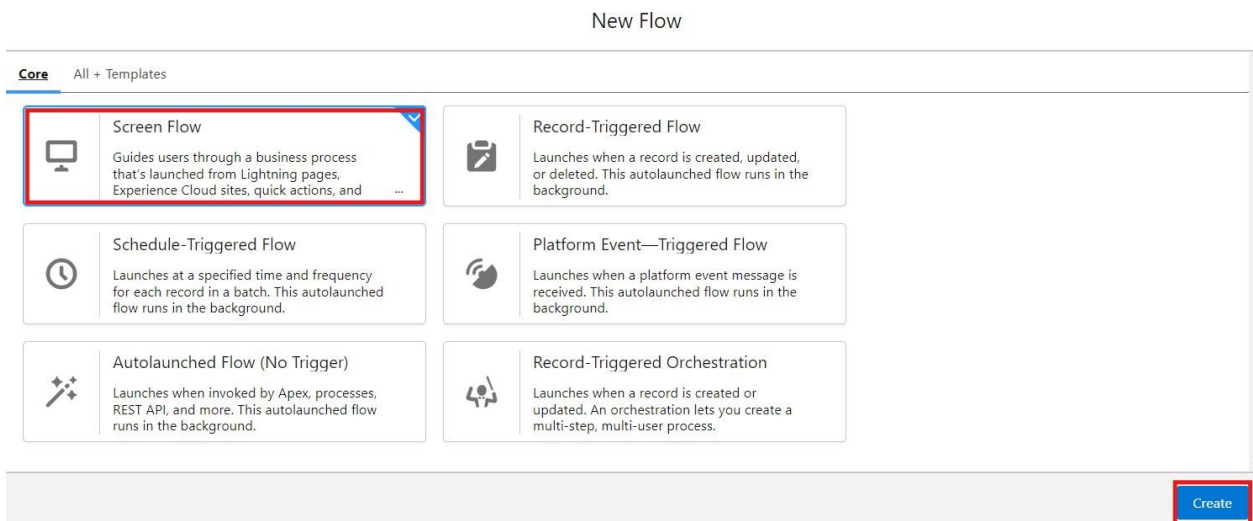
## Milestone 7 : Flows

### Activity 1 : Screen Flow

1. Go to setup, Type Flow in quick find box → Click on the Flow and Select the New Flow.



2. Select Screen Flow and click Create.



3. Click (+) Icon on the flow builder and Select **Screen Component**.  
**Label** : Players screen  
**API** : Players\_screen  
Select the **Picklist** component and place it on the screen.  
**Label** : Team  
**API** : Team  
From Component Pane on the left side select **Component Type** : Picklist  
**Data Type** : Text  
**Choice** : Click New Choice Resources

## Resource Type : Picklist Choice Set

\*Resource Type

Select...

- Choice  
Create a choice option to be used in a screen component.
- Collection Choice Set  
Generate a set of choices from a collection of records.
- Record Choice Set  
Generate a set of choices using a filtered list of records.
- Picklist Choice Set**  
Generate a set of choices by using the values of a picklist or multi-select picklist field.

Cancel Done

**API Name :** Team

**Object :** Player

**Data Type :** Picklist

**Field :** Team

Click Done.

### New Choice Resource

\*Resource Type  
Picklist Choice Set

\*API Name  
Team

Description

\*Object  
Player

\*Data Type  
Picklist

\*Field  
Team\_\_c

Sort Order  
Default Order of Field

Cancel Done

- Click (+) Icon on the flow builder and Select **Get Records Component**.

**Label :** Players

**API Name :** Players

**Object :** Player

**Filter player records :**

**Field :** Team

**Operator :** Equals

**Value** : Click New Resource

**Resource Type** : Variable

**API Name** : Dynamic

**Data Type** : Text

**Default Value** : Screen component : Team (Picklist)

Filter Player Records

Condition Requirements

All Conditions Are Met (AND) ▼

Field Operator Value

Team\_\_c Equals dynamic ×

+ Add Condition

5. Click (+) Icon on the flow builder and Select **Screen Component**.

**Label** : Player\_info

**API Name** : Player\_info

From Component Pane on the left side select **Component Type** : Data table

**API Name** : Player\_info\_table

**Label** : Data Table

**Configure Data Source** : **Source Collection** : Players from players (Player\_\_c)

Edit Screen

Components Fields

Data

Input (1)

Data Table

players flow

Data Table

0 items

Pro... Pla... Tot... AV... Na... Bat... Bo... Ma... Nu... Nu...

No items to display.

Pause Previous Finish

Get more on the AppExchange

Data Table

Label

Data Table

Use Label as the table title

Configure Data Source

Source Collection

Enter value or search resources...

New Resource

RECORD COLLECTION VARIABLES

Players from Players

Player\_\_c

SCREEN COMPONENTS

Player\_info\_table

Data Table

**In Configure Columns** : Add all the required Columns

Profile Picture | Player Name | Total Runs | AVG Score | Batting Style | Bowling

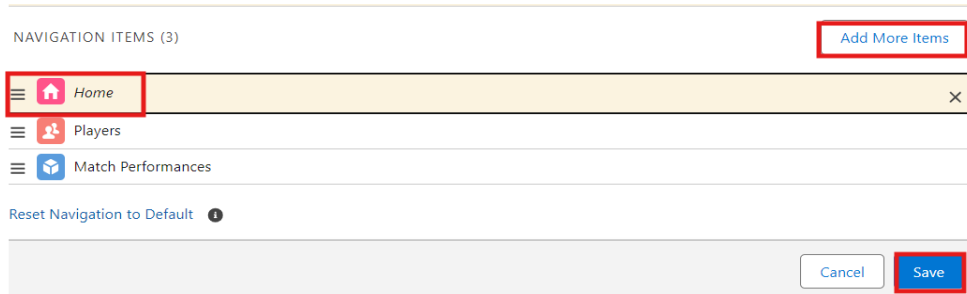
Style | Matches Played | Number of 50's | Number of 100's

Click Done.

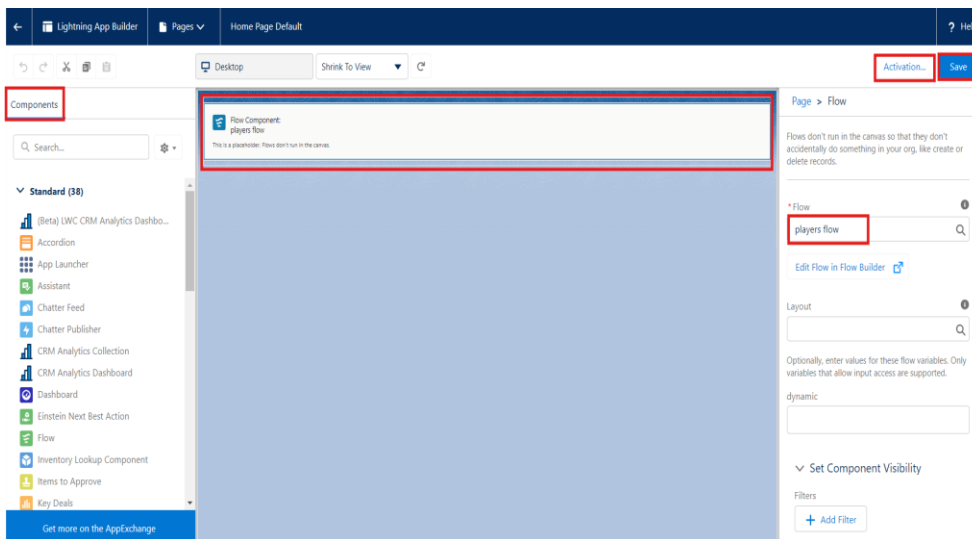


## Activity 2 : Flow on Home Page

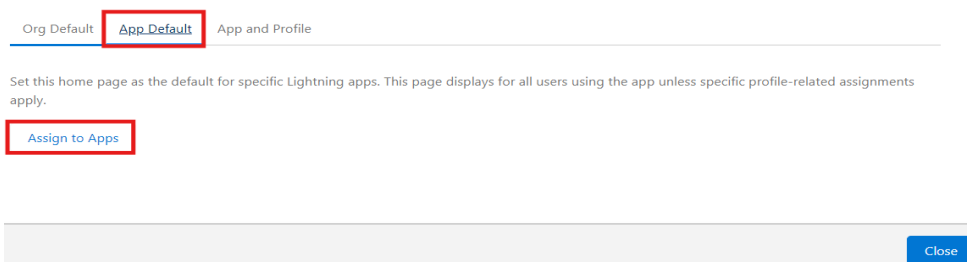
1. On your CRICKSTATS Application on the top right of the Navigation Bar click on **Personalize your nav bar icon (Pencil Icon)** and Add Home Page and click Save.



2. Remove all other components and Add **FLOW** component from the standard components on the left side.
3. **Flow** : Players Flow



4. Click **Save** and **Activation**.
5. Click on **App Default** and select **Assign to Apps**.



6. Select **CRICKSTAS** App and Click **Next** and then click **Save**.

Select Apps

Selected Lightning apps display "Home Page Default" as the home page.

| Lightning Apps (12)                               |                                                                           | 1 Selected |
|---------------------------------------------------|---------------------------------------------------------------------------|------------|
| App Name                                          | Description                                                               |            |
| <input type="checkbox"/> Commerce                 | Manage your store's products, catalogs, and pricebooks.                   |            |
| <input checked="" type="checkbox"/> CRICKSTAS     | An application to collect, manage, and analyze cricket players' perfor... |            |
| <input type="checkbox"/> Data Manager             | Use Data Manager to view limits, monitor usage, and manage recipes.       |            |
| <input type="checkbox"/> LightningBolt            | Discover and manage business solutions designed for your industry.        |            |
| <input type="checkbox"/> LightningInstrumentation | View Adoption and Usage Metrics for Lightning Experience                  |            |
| <input type="checkbox"/> Queue Management         | Create and manage queues for your business.                               |            |

[Cancel](#) [Back](#) [Next](#)

## 7. FUNCTIONAL & PERFORMANCE TESTING

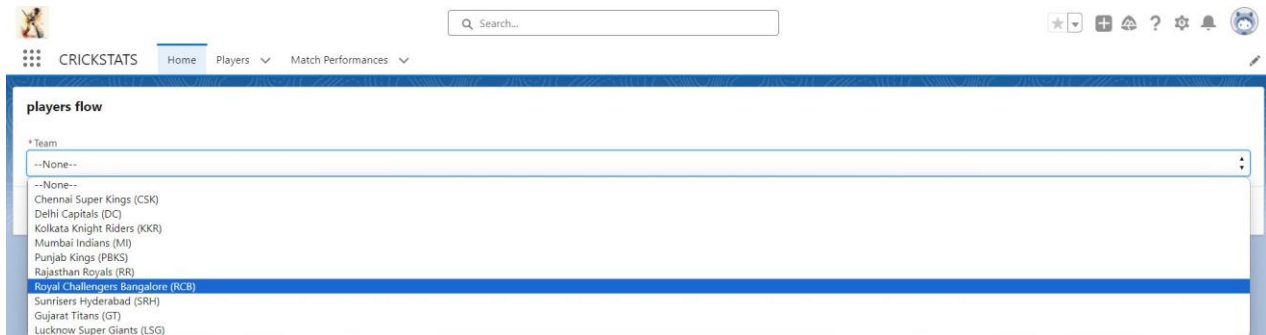
During the testing phase, screenshots were captured from the Salesforce Lightning user interface to verify the correct functioning of all implemented features in the CRICKSTATS system. These include record creation for Player and Match Performance objects; validation rule checks for mandatory fields, score limits, and input constraints; verification of record type behavior for Domestic and International match categories; and flow execution for automatic updating of player statistics and data consistency. The functionality of roll-up summary fields was also tested to ensure accurate calculation of total runs, wickets, averages, and performance metrics.

In addition, system performance was evaluated by monitoring response time during data entry, report generation, and dashboard loading. The application was tested under multiple user interactions to ensure stable performance without delays. Screenshots of generated reports and interactive dashboards were documented to confirm accurate data visualization, real-time updates, and reliable performance tracking. These testing

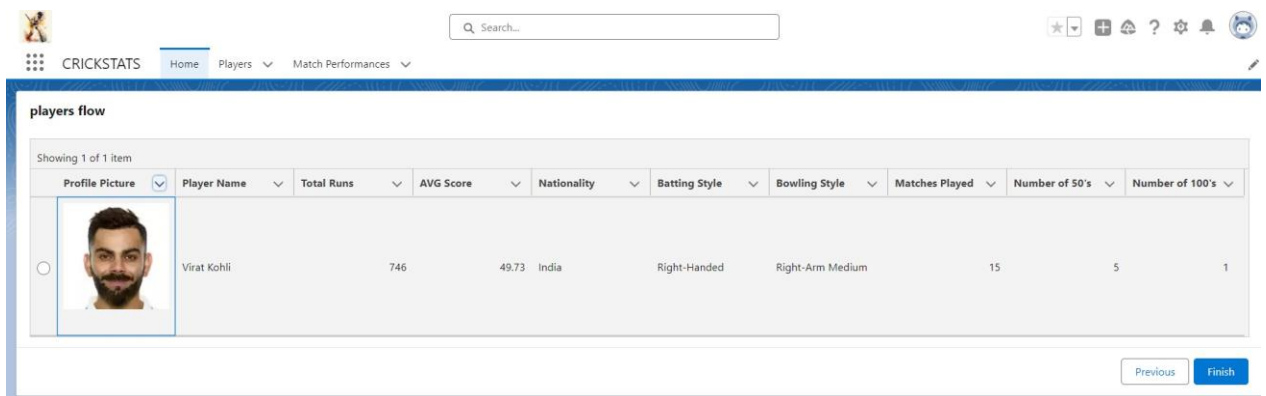
activities ensured that the CRICKSTATS platform met both functional and performance requirements and was ready for deployment.

**NOTE :** Enter some records in the Player Object and Match Performance Object to get the results.

1. Select one Team from the Picklist and Click **Next**.

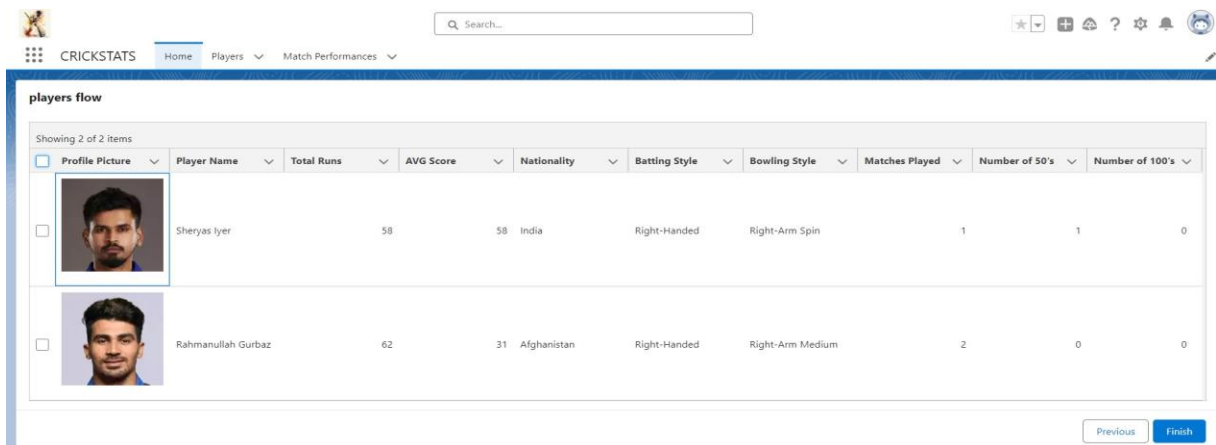


The screenshot shows the 'players flow' section of the CRICKSTATS application. At the top, there is a search bar and navigation links for Home, Players, and Match Performances. Below the navigation bar, a dropdown menu is open for selecting a team. The dropdown list includes the following options: --None--, Chennai Super Kings (CSK), Delhi Capitals (DC), Kolkata Knight Riders (KKR), Mumbai Indians (MI), Punjab Kings (PBKS), Rajasthan Royals (RR), **Royal Challengers Bangalore (RCB)** (highlighted), Sunrisers Hyderabad (SRH), Gujarat Titans (GT), and Lucknow Super Giants (LSG).



The screenshot shows the 'players flow' section of the CRICKSTATS application. The table displays 1 item. The table has the following columns: Profile Picture, Player Name, Total Runs, AVG Score, Nationality, Batting Style, Bowling Style, Matches Played, Number of 50's, and Number of 100's.

| Profile Picture                                                                     | Player Name | Total Runs | AVG Score | Nationality | Batting Style | Bowling Style    | Matches Played | Number of 50's | Number of 100's |
|-------------------------------------------------------------------------------------|-------------|------------|-----------|-------------|---------------|------------------|----------------|----------------|-----------------|
|  | Virat Kohli | 746        | 49.73     | India       | Right-Handed  | Right-Arm Medium | 15             | 5              | 1               |



The screenshot shows the 'players flow' section of the CRICKSTATS application. The table displays 2 items. The table has the following columns: Profile Picture, Player Name, Total Runs, AVG Score, Nationality, Batting Style, Bowling Style, Matches Played, Number of 50's, and Number of 100's.

| Profile Picture                                                                     | Player Name        | Total Runs | AVG Score | Nationality | Batting Style | Bowling Style    | Matches Played | Number of 50's | Number of 100's |
|-------------------------------------------------------------------------------------|--------------------|------------|-----------|-------------|---------------|------------------|----------------|----------------|-----------------|
|  | Shreyas Iyer       | 58         | 58        | India       | Right-Handed  | Right-Arm Spin   | 1              | 1              | 0               |
|  | Rahmanullah Gurbaz | 62         | 31        | Afghanistan | Right-Handed  | Right-Arm Medium | 2              | 0              | 0               |

## **8. ADVANTAGES & DISADVANTAGES**

### **Advantages**

- Centralized system
- Real-time analytics
- Secure platform
- Scalable design
- Reduced manual work

### **Disadvantages**

- Requires Salesforce license
- Internet dependency
- Limited offline access

## **9. Conclusion:**

CRICKSTATS successfully implements a cloud-based cricket performance tracking system using Salesforce. The system automates data management, improves reporting accuracy, and supports informed decision-making.

The project demonstrates practical application of SDLC, Agile methodology, and CRM-based development.

## **10. FUTURE SCOPE**

- Mobile application integration
- AI-based performance prediction

- Video analysis integration
- Player ranking system
- Public statistics portal
- Multi-league support

## **11. APPENDIX**

- Salesforce Objects & Fields
- Manual match statistics
- Flow Configurations

**Thank You**